

Data Dictionary

Zero-Inflated Poisson (ZIP) Latent-Class Trajectories

File: zip_poisson.sas

Last updated: July 21, 2026

Package: Traj2 (prototype, under QA)

1. Overview

This file fits zero-inflated Poisson latent-class (growth-mixture) trajectory models in PROC NLMIXED, for count outcomes with excess zeros (e.g. quarterly ED visits). Each class has a log-link Poisson mean trajectory and a logit zero-inflation trajectory. Starting values are set via %let lines; unset parameters default to 0. The zero-mass probability uses a numerically stable log-sum-exp evaluation.

2. Datasets

Name	Description
sim_long / sim_wide	Long / wide outputs of the bounded-count simulator (%sim_data).
BASE_FILE_SRS	Wide input: BENE_ID, SUM_Q1 . . SUM_Q12, quar1 . . quar12 (values 1..T).
pe_zip	ParameterEstimates from the fit (default name).
fit_zip	FitStatistics from the fit (default name).
traj / mix	Per-class ZIP mean curves and the mixture-mean series (plotting).

3. Macro Arguments (%ct_zip_pois_nlmixed)

Name	Type / Default	Description
data	name, BASE_FILE_SRS	Input wide-format dataset.
id	name, BENE_ID	Subject identifier.
yvars	varlist, SUM_Q1-SUM_Q12	Repeated-measure count outcomes.
nclass	integer, 6	Number of latent classes K.
class_labels	list, A B C D E F	Space-separated labels; must match nclass.
order	integer, 2	Mean-curve polynomial degree.
p_order	integer, 0	Zero-inflation logit polynomial degree.
T	integer, 12	Number of time points.
tech	name, newrap	Optimization technique.
maxiter	integer, 500	Maximum iterations.
pe_out	name, pe_zip	Output dataset for parameter estimates.
fit_out	name, fit_zip	Output dataset for fit statistics.
bounds	text, optional	Parameter bounds for stability.

4. Key Variables and Parameters

Name	Type	Description
BENE_ID	Num	Subject identifier.
Y[T]	Array	Count outcome over time inside the model.
alpha0_[label]	Num	Class-membership logit (class A is reference, weight 1/den).
beta0/1/2/3_[label]	Num	Log-link Poisson mean polynomial coefficients.
gamma0/1/2/3_[label]	Num	Zero-inflation logit polynomial coefficients.
mu_[label][T]	Array	Class mean count trajectory, $\exp(\eta)$.
logitp_[label][T]	Array	Class zero-inflation logit over time.
p_[label][T]	Array	Class structural-zero probability over time.
pi_[label][T]	Array	Per-time log-likelihood contribution.
w_[label]	Num	Class mixing weight (softmax of alphas).
prod_[label]	Num	Summed per-class log-likelihood across time.
ll	Num	Mixture log-likelihood (log-sum-exp) for the model.

5. Starting-Value Macro Variables (Section [C])

Name	Description
alpha0_B, alpha0_C, ...	Mixing logits; class A has none (reference).
beta0_A, beta1_A, ...	Mean-curve coefficients per class.
gamma0_A, gamma0_B, ...	Zero-inflation logits per class.
(any unset parameter)	Defaults to 0 in the PARMS statement.

6. Macros

Name	Purpose
%sim_data	Bounded-count simulator (poisson, BIN4, TPOIS4, CAT5).
%emit_parm	Emits one PARMS entry from a %let value or a 0 default.
%ct_array_from_list	Declares the outcome array.
%ct_declare_mu_arrays / _pi_ / _zip_	Declare mu, pi, logitp/p arrays per class.
%ct_parms_all	Builds the single PARMS statement (alphas, betas, gammas).
%ct_fill_mu_poly	Fills class mean trajectories.
%ct_fill_zip_poly	Fills class zero-inflation trajectories.

Name	Purpose
%ct_accumulate_zip_ll	Accumulates per-class ZIP log-likelihood.
%ct_mixture_ll_zip	Mixes classes via softmax weights (log-sum-exp).
%ct_zip_pois_nlmixed	Main fit macro.
%ct_zip_plots	Class curves and mixture-mean overlay.

7. Notes and Assumptions

- Prototype: interfaces and behavior may change before release.
- Class 1 (label A) is the mixing reference; no α_0 for it.
- Exactly one PARMS statement is built; no carryover between runs.
- Missing Y values are skipped in the likelihood.
- Model selection by BIC (in `fit_zip`) across candidate K.